

6

Inventories

Learning Objectives

1

Discuss how to classify and determine inventory.

2

Apply inventory cost flow methods and discuss their financial effects.

3

Indicate the effects of inventory errors on the financial statements.

4

Explain the statement presentation and analysis of inventory.

Classifying Inventory

Merchandising Company

One Classification:

- ◆ **Inventory**

Helpful Hint

Regardless of the classification, companies report all inventories under Current Assets on the balance sheet.

Manufacturing Company

Three Classifications:

- ◆ **Raw Materials**
- ◆ **Work in Process**
- ◆ **Finished Goods**

Accounting Across the Organization Ford



© PeskyMonkey/iStockphoto

A Big Hiccup

JIT can save a company a lot of money, but it isn't without risk. An unexpected disruption in the supply chain can cost a company a lot of money. Japanese automakers experienced just such a disruption when a 6.8-magnitude earthquake caused major damage to the company that produces 50% of their piston rings. The rings themselves cost only \$1.50, but you can't

make a car without them. As a result, the automakers were forced to shut down production for a few days—a loss of tens of thousands of cars.

Similarly, a major snowstorm halted production at the Canadian plants of **Ford**. A Ford spokesperson said, "Because the plants run with just-in-time inventory, we don't have large stockpiles of parts sitting around. When you have a somewhat significant disruption, you can pretty quickly run out of parts."

Sources: Amy Chozick, "A Key Strategy of Japan's Car Makers Backfires," *Wall Street Journal* (July 20, 2007); and Kate Linebaugh, "Canada Military Evacuates Motorists Stranded by Snow," *Wall Street Journal* (December 15, 2010).

What steps might the companies take to avoid such a serious disruption in the future? (Go to **WileyPLUS** for this answer and additional questions.)

Determining Inventory Quantities

Physical Inventory taken for two reasons:

Perpetual System

1. Check accuracy of inventory records.
2. Determine amount of inventory lost due to wasted raw materials, shoplifting, or employee theft.

Periodic System

3. Determine the inventory on hand.
4. Determine the cost of goods sold for the period.

Determining Inventory Quantities

TAKING A PHYSICAL INVENTORY

Involves counting, weighing, or measuring each kind of inventory on hand.

Companies often “take inventory”

- ◆ when the business is closed or business is slow.
- ◆ at the end of the accounting period.

ETHICS NOTE

In a famous fraud, a salad oil company filled its storage tanks mostly with water. The oil rose to the top, so auditors thought the tanks were full of oil. The company also said it had more tanks than it really did: It repainted numbers on the tanks to confuse auditors.

Falsifying Inventory to Boost Income

Managers at women's apparel maker **Leslie Fay** were convicted of falsifying inventory records to boost net income—and consequently to boost management bonuses. In another case, executives at **Craig Consumer Electronics** were accused of defrauding

lenders by manipulating inventory records. The indictment said the company classified “defective goods as new or refurbished” and claimed that it owned certain shipments “from overseas suppliers” when, in fact, Craig either did not own the shipments or the shipments did not exist.

What effect does an overstatement of inventory have on a company's financial statements? (Go to **WileyPLUS** for this answer and additional questions.)



© Greg Brookes/iStockphoto

Determining Inventory Quantities

DETERMINING OWNERSHIP OF GOODS

GOODS IN TRANSIT

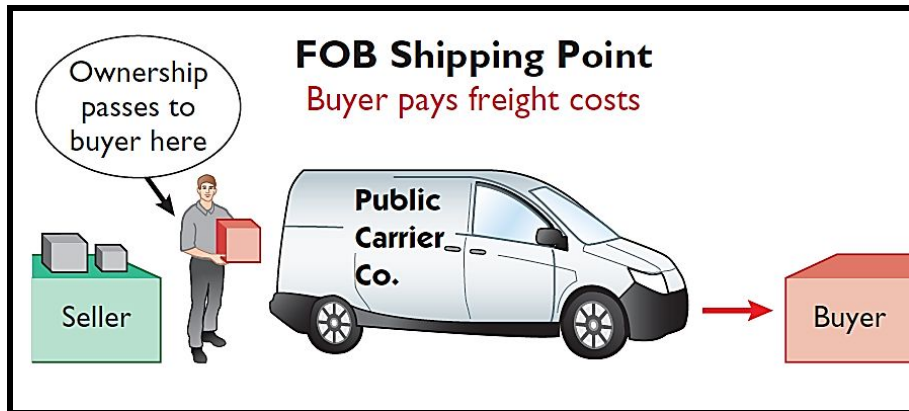
- ◆ Purchased goods not yet received.
- ◆ Sold goods not yet delivered.

Goods in transit should be included in the inventory of the company that has **legal title** to the goods. Legal title is determined by the **terms of sale**.

Determining Ownership of Goods

GOODS IN TRANSIT

Illustration 6-2
Terms of sale



Ownership of the goods passes to the buyer when the public carrier accepts the goods from the seller.

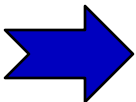


Ownership of the goods remains with the seller until the goods reach the buyer.

Determining Ownership of Goods

Question

Goods in transit should be included in the inventory of the buyer when the:

- a. public carrier accepts the goods from the seller.
- b. goods reach the buyer.
- c. terms of sale are FOB destination.
-  d. terms of sale are FOB shipping point.

Determining Ownership of Goods

CONSIGNED GOODS

To hold the goods of other parties and try to sell the goods for them for a fee, but without taking ownership of the goods.

Many car, boat, and antique dealers sell goods on consignment, why?

ANATOMY OF A FRAUD

Ted Nickerson, CEO of clock manufacturer Dally Industries, was feared by all of his employees. Ted also had expensive tastes. To support this habit, Ted took out large loans, which he collateralized with his shares of Dally Industries stock. If the price of Dally's stock fell, he was required to provide the bank with more shares of stock. To achieve target net income figures and thus maintain the stock price, Ted coerced employees in the company to alter inventory figures. Inventory quantities were manipulated by changing the amounts on inventory control tags after the year-end physical inventory count. For example, if a tag said there were 20 units of a particular item, the tag was changed to 220. Similarly, the unit costs that were used to determine the value of ending inventory were increased from, for example, \$125 per unit to \$1,250. Both of these fraudulent changes had the effect of increasing the amount of reported ending inventory. This reduced cost of goods sold and increased net income.

Total take: \$245,000

Source: Adapted from Wells, *Fraud Casebook* (2007), pp. 502–509.

DO IT!

1

Rules of Ownership

Hasbeen Company completed its inventory count. It arrived at a total inventory value of \$200,000. You have been given the information listed below. Discuss how this information affects the reported cost of inventory.

1. Hasbeen included in the inventory goods held on consignment for Falls Co., costing \$15,000.
2. The company did not include in the count purchased goods of \$10,000, which were in transit (terms: FOB shipping point).
3. The company did not include in the count inventory that had been sold with a cost of \$12,000, which was in transit (terms: FOB shipping point).

Solution

1. Goods of \$15,000 held on consignment should be deducted from the inventory count.
2. The goods of \$10,000 purchased FOB shipping point should be added to the inventory count.
3. Item 3 was treated correctly.

Inventory should be \$195,000
(\$200,000 - \$15,000 + \$10,000).

Inventory is accounted for at cost.

- ◆ Cost includes all expenditures necessary to acquire goods and place them in a condition ready for sale.
- ◆ Unit costs are applied to quantities to compute the total cost of the inventory and the cost of goods sold using the following costing methods:
 - ▶ Specific identification
 - ▶ First-in, first-out (FIFO)
 - ▶ Last-in, first-out (LIFO)
 - ▶ Average-cost

**Cost Flow
Assumptions**

Inventory Costing

Illustration: Crivitz TV Company purchases three identical 50-inch TVs on different dates at costs of \$700, \$750, and \$800. During the year Crivitz sold two sets at \$1,200 each. These facts are summarized below.

Illustration 6-3
Data for inventory
costing example

Purchases

February 3 1 TV at \$700

March 5 1 TV at \$750

May 22 1 TV at \$800

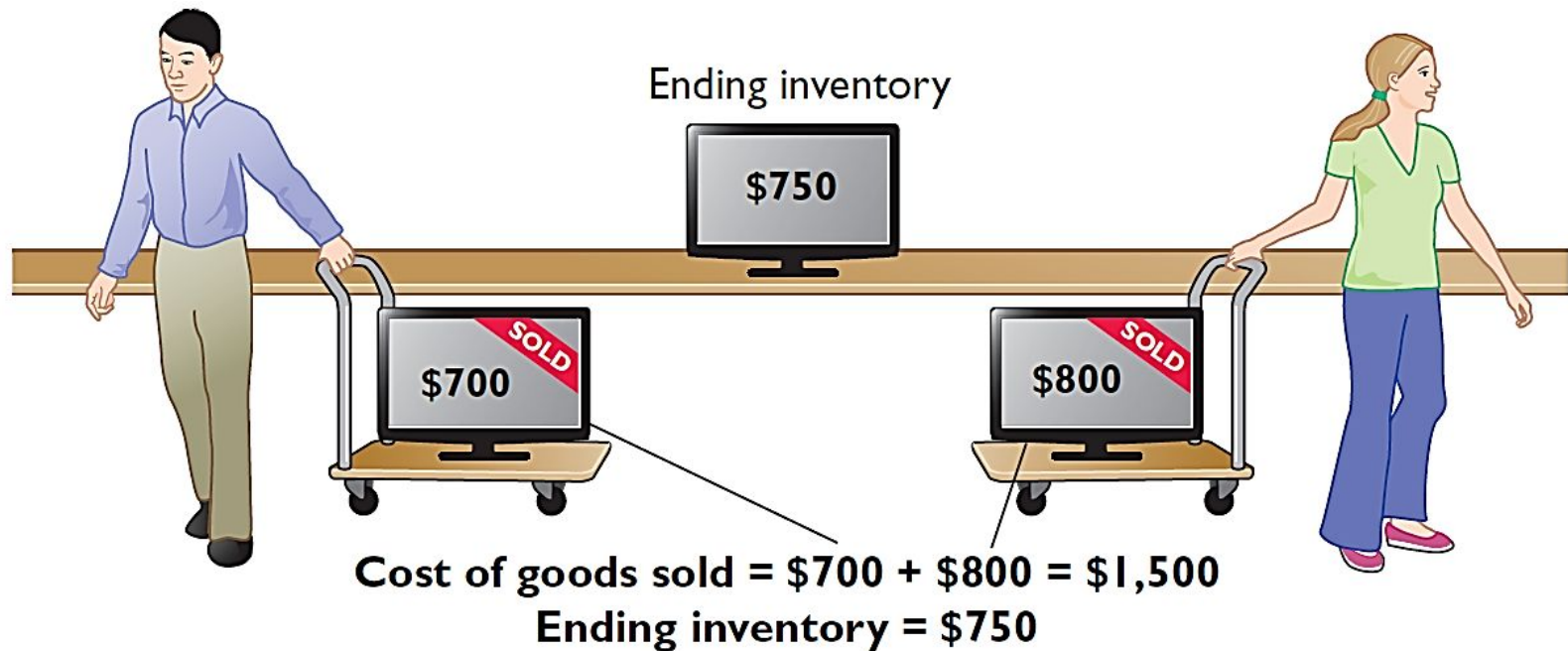
Sales

June 1 2 TVs for \$2,400 ($\$1,200 \times 2$)

Specific Identification

If Crivitz sold the TVs it purchased on February 3 and May 22, then its cost of goods sold is \$1,500 (\$700 + \$800), and its ending inventory is \$750.

Illustration 6-4



Specific Identification

Actual physical flow costing method in which items still in inventory are specifically costed to arrive at the total cost of the ending inventory.

- ◆ Practice is relatively rare.
- ◆ Most companies make assumptions (**cost flow assumptions**) about which units were sold.

ETHICS NOTE

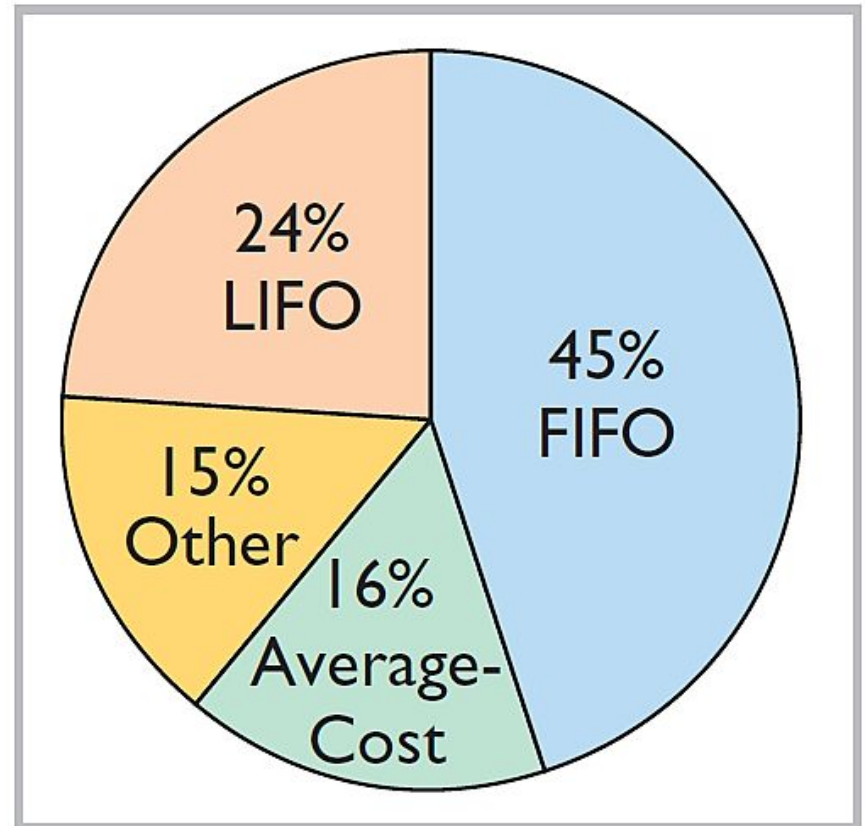
A major disadvantage of the specific identification method is that management may be able to manipulate net income. For example, it can boost net income by selling units purchased at a low cost, or reduce net income by selling units purchased at a high cost.

Cost Flow Assumptions

Cost flow assumptions

DO NOT need to be
consistent with the
physical movement of
the goods

Illustration 6-12
Use of cost flow methods in
major U.S. companies



Cost Flow Assumptions

Illustration: Data for Houston Electronics' Astro condensers.

Illustration 6-5

HOUSTON ELECTRONICS Astro Condensers				
<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total units available for sale	1,000		<u>\$12,000</u>
	Units in ending inventory	<u>450</u>		
	Units sold	<u>550</u>		

(Beginning Inventory + Purchases) - Ending Inventory = Cost of Goods Sold

Cost Flow Assumptions

FIRST-IN, FIRST-OUT (FIFO)

- ◆ **Costs of the earliest goods purchased** are the first to be recognized in determining cost of goods sold.
- ◆ Often parallels actual physical flow of merchandise.
- ◆ Companies determine the cost of the ending inventory by taking the unit cost of the most recent purchase and working backward until all units of inventory have been costed.

FIRST-IN, FIRST-OUT (FIFO)

Illustration 6-6 ■

COST OF GOODS AVAILABLE FOR SALE

<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	<u>1,000</u>		<u>\$12,000</u>

STEP 1: ENDING INVENTORY

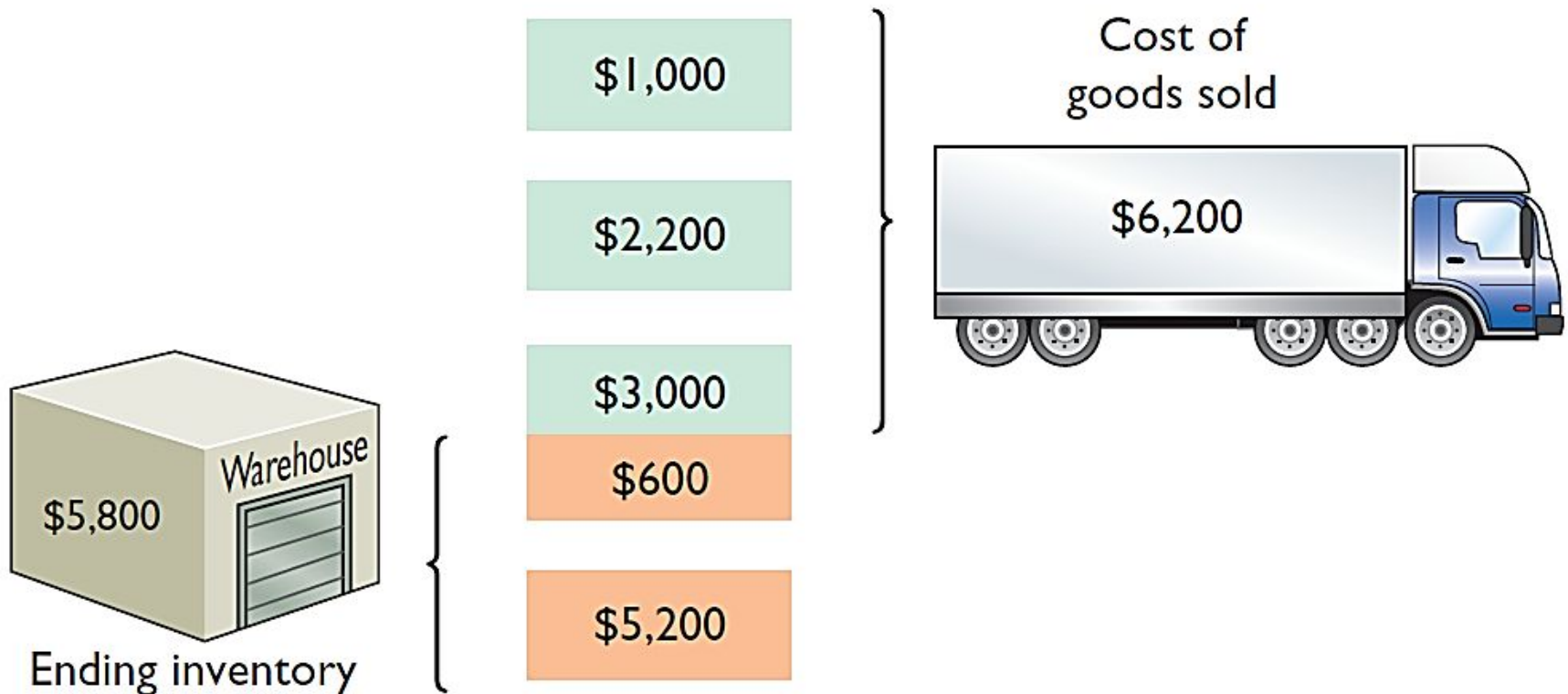
STEP 2: COST OF GOODS SOLD

<u>Date</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
-------------	--------------	------------------	-------------------

Cost of goods available for sale \$12,000

FIRST-IN, FIRST-OUT (FIFO)

Illustration 6-6 ■



Helpful Hint Another way of thinking about the calculation of FIFO ending inventory is the **LISH assumption**—last in still here.

Cost Flow Assumptions

LAST-IN, FIRST-OUT (LIFO)

- ◆ **Costs of the latest goods purchased** are the first to be recognized in determining cost of goods sold.
- ◆ **Seldom coincides** with actual physical flow of merchandise.
- ◆ **Exceptions** include goods stored in piles, such as coal or hay.

LAST-IN, FIRST-OUT (LIFO)

Illustration 6-8 ■

COST OF GOODS AVAILABLE FOR SALE

<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	<u>1,000</u>		<u>\$12,000</u>

STEP 1: ENDING INVENTORY

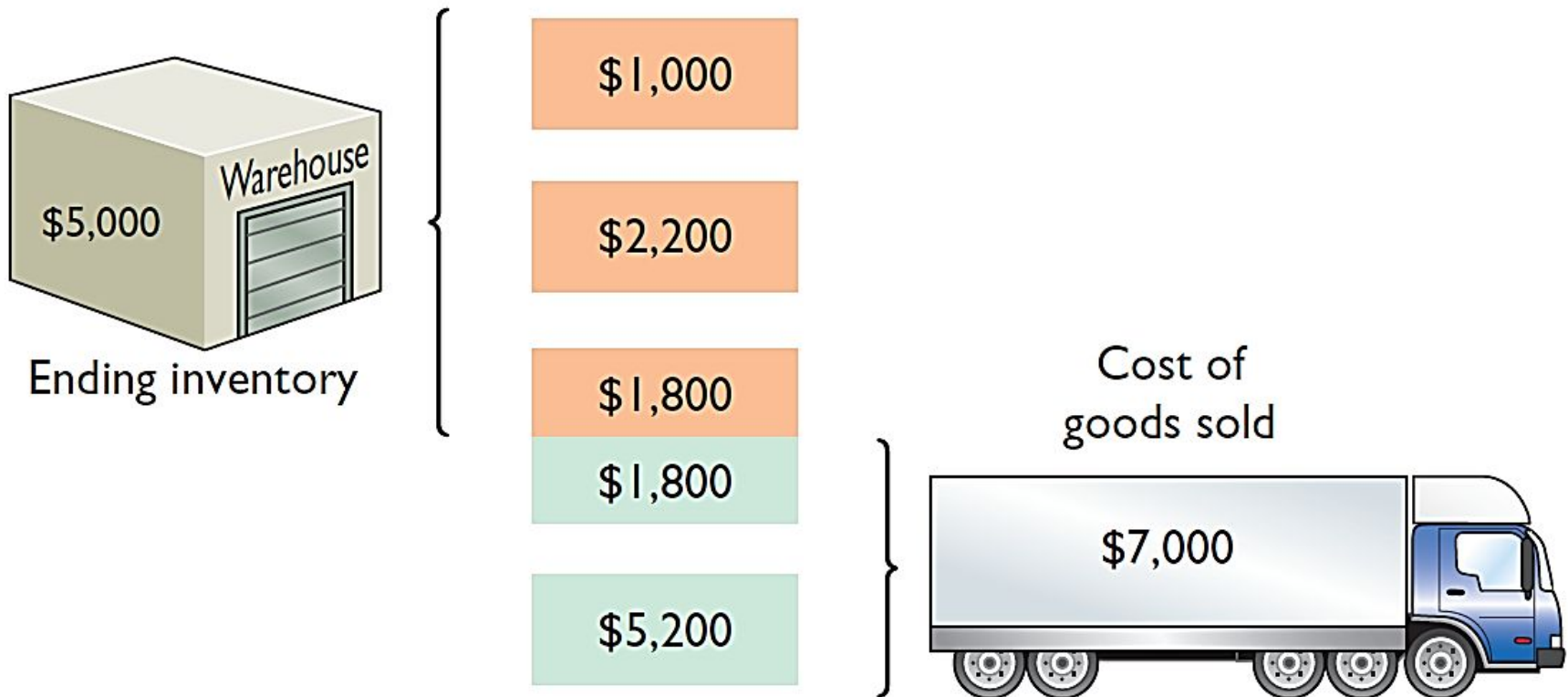
STEP 2: COST OF GOODS SOLD

<u>Date</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
-------------	--------------	------------------	-------------------

Cost of goods available for sale	\$12,000
----------------------------------	----------

LAST-IN, FIRST-OUT (LIFO)

Illustration 6-8 ■



Helpful Hint Another way of thinking about the calculation of LIFO **ending inventory** is the FISH assumption—first in still here.

Cost Flow Assumptions

AVERAGE-COST

- ◆ Allocates cost of goods available for sale on the basis of **weighted-average unit cost** incurred.
- ◆ Applies weighted-average unit cost to the **units on hand** to determine cost of the ending inventory.

AVERAGE-COST

Illustration 6-11 ■

COST OF GOODS AVAILABLE FOR SALE

<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>
Jan. 1	Beginning inventory	100	\$10	\$ 1,000
Apr. 15	Purchase	200	11	2,200
Aug. 24	Purchase	300	12	3,600
Nov. 27	Purchase	400	13	5,200
	Total	<u>1,000</u>		<u>\$12,000</u>

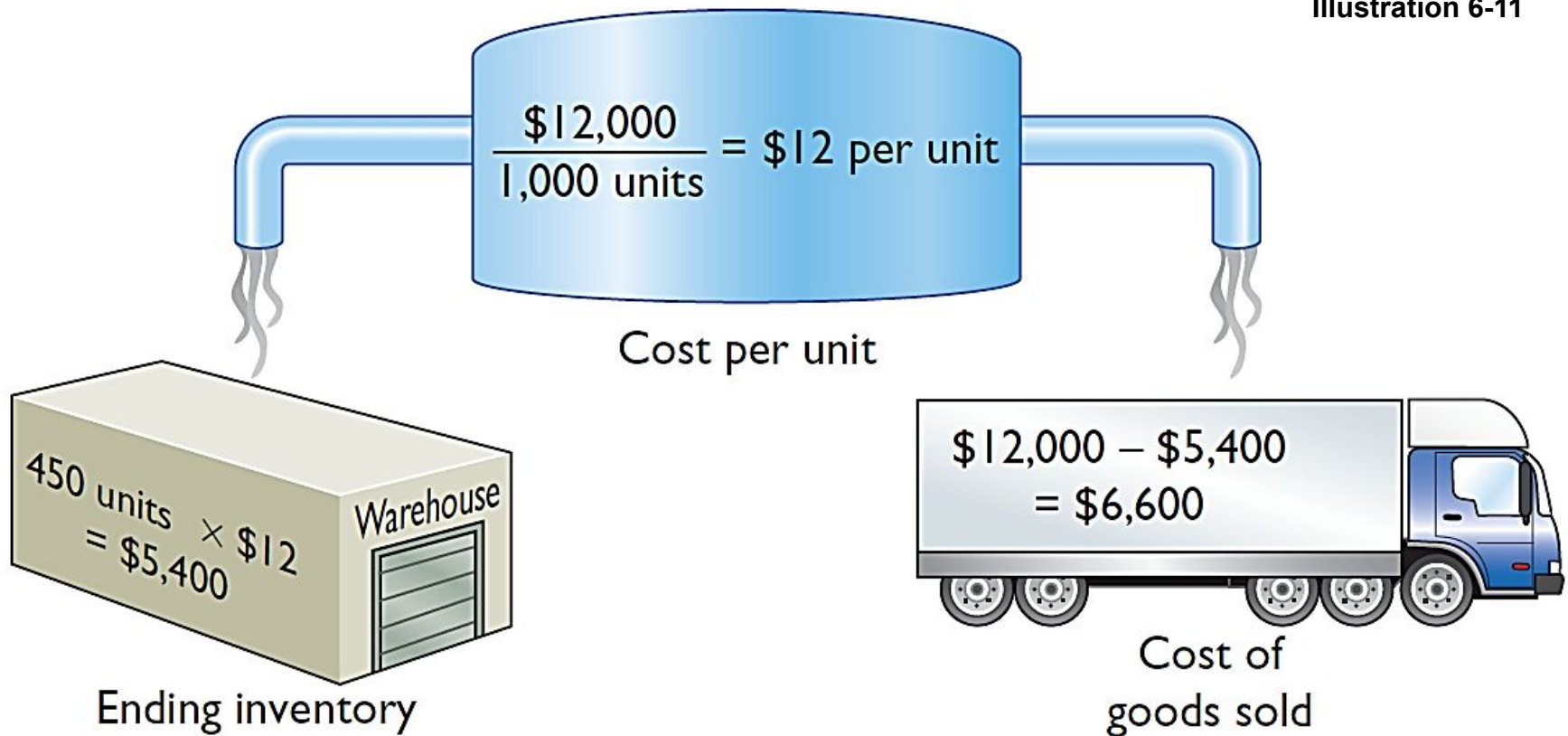
STEP 1: ENDING INVENTORY

STEP 2: COST OF GOODS SOLD

\$12,000	÷	1,000	=	\$12	Cost of goods available for sale	\$12,000
<u>Units</u>		<u>Unit Cost</u>		<u>Total Cost</u>		

AVERAGE-COST

Illustration 6-11



Inventory Costing

Financial Statement and Tax Effects of Cost Flow Methods

Each of the three cost flow methods is acceptable for use.

- ◆ **Reebok International Ltd.** and **Wendy's International** currently use the FIFO method.
- ◆ **Campbell Soup Company**, **Krogers**, and **Walgreen Drugs** use LIFO for part or all of their inventory.
- ◆ **Bristol-Myers Squibb**, **Starbucks**, and **Motorola** use the average-cost method.
- ◆ **Stanley Black & Decker Manufacturing Company** uses LIFO for domestic inventories and FIFO for foreign inventories.

Financial Statement and Tax Effects

INCOME STATEMENT EFFECTS

Illustration 6-13
Comparative effects of
cost flow methods

HOUSTON ELECTRONICS Condensed Income Statements			
	FIFO	LIFO	Average-Cost
Sales revenue	\$18,500	\$18,500	\$18,500
Beginning inventory	1,000	1,000	1,000
Purchases	11,000	11,000	11,000
Cost of goods available for sale	12,000	12,000	12,000
Ending inventory	5,800	5,000	5,400
Cost of goods sold	6,200	7,000	6,600
Gross profit	12,300	11,500	11,900
Operating expenses	9,000	9,000	9,000
Income before income taxes	3,300	2,500	2,900
Income tax expense (30%)	990	750	870
Net income	\$ 2,310	\$ 1,750	\$ 2,030

Financial Statement and Tax Effects

BALANCE SHEET EFFECTS

- ◆ A major **advantage** of the **FIFO method** is that in a period of inflation, the costs allocated to ending inventory will **approximate their current cost**.
- ◆ A major **shortcoming** of the **LIFO method** is that in a period of inflation, the costs allocated to ending inventory may be **significantly understated** in terms of current cost.

Financial Statement and Tax Effects

TAX EFFECTS

- ◆ Both inventory and net income are higher when companies use **FIFO** in a period of inflation.
- ◆ **LIFO** results in the lowest income taxes (because of lower net income) during times of rising prices.

Helpful Hint

A tax rule, often referred to as the LIFO conformity rule, requires that if companies use LIFO for tax purposes they must also use it for financial reporting purposes.

Inventory Costing

Using Cost Flow Methods Consistently

- ◆ Method should be used consistently, enhances comparability.
- ◆ Although consistency is preferred, a company may change its inventory costing method.

Illustration 6-14

Disclosure of change in cost flow method

Real
World

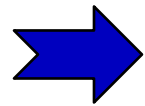
QUAKER OATS Notes to the Financial Statements

Note 1: Effective July 1, the Company adopted the LIFO cost flow assumption for valuing the majority of U.S. Grocery Products inventories. The Company believes that the use of the LIFO method better matches current costs with current revenues. The effect of this change on the current year was to decrease net income by \$16.0 million.

Cost Flow Assumptions

Question

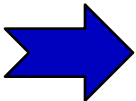
The cost flow method that often parallels the actual physical flow of merchandise is the:

-  a. FIFO method.
- b. LIFO method.
- c. average cost method.
- d. gross profit method.

Cost Flow Assumptions

Question

In a period of inflation, the cost flow method that results in the lowest income taxes is the:

- a. FIFO method.
-  b. LIFO method.
- c. average cost method.
- d. gross profit method.

Is LIFO Fair?

ExxonMobil Corporation, like many U.S. companies, uses LIFO to value its inventory for financial reporting and tax purposes. In one recent year, this resulted in a cost of goods sold figure that was \$5.6 billion higher than under FIFO. By increasing cost of goods sold, ExxonMobil reduces net income, which reduces taxes. Critics say that LIFO provides an unfair “tax dodge.” As Congress looks for more sources of

tax revenue, some lawmakers favor the elimination of LIFO. Supporters of LIFO argue that the method is conceptually sound because it matches current costs with current revenues. In addition, they point out that this matching provides protection against inflation.

International accounting standards do not allow the use of LIFO. Because of this, the net income of foreign oil companies such as **BP** and **Royal Dutch Shell** are not directly comparable to U.S. companies, which makes analysis difficult.

Source: David Reilly, “Big Oil’s Accounting Methods Fuel Criticism,” Wall Street Journal (August 8, 2006), p. C1.

What are the arguments for and against the use of LIFO?
(Go to **WileyPLUS** for this answer and additional questions.)

DO IT!

2

Cost Flow Methods

The accounting records of Shumway Ag Implements show the following data.

Beginning inventory	4,000 units at \$ 3
Purchases	6,000 units at \$ 4
Sales	7,000 units at \$12

Determine the cost of goods sold during the period under a periodic inventory system using (a) the FIFO method, (b) the LIFO method, and (c) the average-cost method.

Solution

Cost of goods available for sale =

Ending inventory =

(a) FIFO:

(b) LIFO:

(c) Average cost per unit:

Average-cost:

Common Cause:

- ◆ Failure to count or price inventory correctly.
- ◆ Not properly recognizing the transfer of legal title to goods in transit.
- ◆ Errors affect both the income statement and balance sheet.

Income Statement Effects

Inventory errors affect the computation of cost of goods sold and net income **in two periods**.

Illustration 6-15

$$\begin{array}{ccccccc} \text{Beginning} & & & & & & \\ \text{Inventory} & + & \text{Cost of} & - & \text{Ending} & = & \text{Cost of} \\ & & \text{Goods} & & \text{Inventory} & & \text{Goods} \\ & & \text{Purchased} & & & & \text{Sold} \end{array}$$

Illustration 6-16

<u>When Inventory Error:</u>	<u>Cost of Goods Sold Is:</u>	<u>Net Income Is:</u>
Understates beginning inventory	Understated	Overstated
Overstates beginning inventory	Overstated	Understated
Understates ending inventory	Overstated	Understated
Overstates ending inventory	Understated	Overstated

Income Statement Effects

Inventory errors affect the computation of cost of goods sold and net income **in two periods**.

- ◆ An error in ending inventory of the current period will have a **reverse effect on net income of the next accounting period**.
- ◆ Over the two years, the total net income is correct because the errors **offset each other**.
- ◆ Ending inventory depends entirely on the accuracy of taking and costing the inventory.

Income Statement Effects

Illustration 6-17
Effects of inventory errors on
two years' income statements

	2016		2017	
	Incorrect	Correct	Incorrect	Correct
Sales	\$ 80,000	\$ 80,000	\$ 90,000	\$ 90,000
Beginning inventory	20,000	20,000	12,000	15,000
Cost of goods purchased	40,000	40,000	68,000	68,000
Cost of goods available	60,000	60,000	80,000	83,000
Ending inventory	12,000	15,000	23,000	23,000
Cost of good sold	48,000	45,000	57,000	60,000
Gross profit	32,000	35,000	33,000	30,000
Operating expenses	10,000	10,000	20,000	20,000
Net income	\$ 22,000	\$ 25,000	\$ 13,000	\$ 10,000

Combined income for
2-year period is correct.

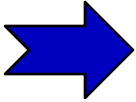
(\$3,000)
Net Income
understated

\$3,000
Net Income
overstated

Income Statement Effects

Question

Understating ending inventory will overstate:

- a. assets.
-  b. cost of goods sold.
- c. net income.
- d. stockholders' equity

Balance Sheet Effects

Effect of inventory errors on the balance sheet is determined by using the basic accounting equation: $\text{Assets} = \text{Liabilities} + \text{Stockholders' Equity}$.

Errors in the ending inventory have the following effects.

<u>Ending Inventory Error</u>	<u>Assets</u>	<u>Liabilities</u>	<u>Stockholders' Equity</u>
Overstated	Overstated	No effect	Overstated
Understated	Understated	No effect	Understated

Illustration 6-18

Effects of ending inventory
errors on balance sheet

DO IT!

3

Inventory Errors

Visual Company overstated its 2016 ending inventory by \$22,000. Determine the impact this error has on ending inventory, cost of goods sold, and stockholders' equity in 2016 and 2017.

Solution

	2016	2017
Ending inventory	\$22,000 overstated	No effect
Cost of goods sold	\$22,000 understated	\$22,000 overstated
Stockholders' equity	\$22,000 overstated	No effect

Presentation

Balance Sheet - Inventory classified as current asset.

Income Statement - Cost of goods sold is subtracted from sales.

There also should be disclosure of the

- 1) major inventory classifications,
- 2) basis of accounting (cost or LCM), and
- 3) costing method (FIFO, LIFO, or average-cost).

Lower-of-Cost-or-Net Realizable Value

When the value of inventory is lower than its cost

- ◆ Companies must “write down” the inventory to its **net realizable value**.
- ◆ Net realizable value: Amount that a company expects to realize (receive from the sale of inventory).
- ◆ Example of **conservatism**.

Lower-of-Cost-or-Net Realizable Value

Illustration: Assume that Ken Tuckie TV has the following lines of merchandise with costs and market values as indicated.

	<u>Units</u>	<u>Cost per Unit</u>	<u>Net Realizable Value per Unit</u>	<u>Lower-of-Cost- or-Market</u>	
Flat-screen TVs	100	\$600	\$550	\$ 55,000	$(\$550 \times 100)$
Satellite radios	500	90	104	45,000	$(\$90 \times 500)$
Blu-ray players	850	50	48	40,800	$(\$48 \times 850)$
CDs	3,000	5	6	15,000	$(\$5 \times 3,000)$
Total inventory				<u>\$155,800</u>	

Illustration 6-20
Computation of
lower-of-cost-or-net
realizable value

Statement Presentation and Analysis

Analysis

Inventory management is a double-edged sword

1. **High Inventory Levels** - may incur high carrying costs (e.g., investment, storage, insurance, obsolescence, and damage).
2. **Low Inventory Levels** – may lead to stock-outs and lost sales.

Analysis

Inventory turnover measures the number of times on average the inventory is sold during the period.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Days in inventory measures the average number of days inventory is held.

$$\text{Days in Inventory} = \frac{\text{Days in Year (365)}}{\text{Inventory Turnover}}$$

Analysis

Illustration: **Wal-Mart** reported in its 2014 annual report a beginning inventory of \$43,803 million, an ending inventory of \$44,858 million, and cost of goods sold for the year ended January 31, 2014, of \$358,069 million. The inventory turnover formula and computation for Wal-Mart are shown below.

Illustration 6-21

Cost of Goods Sold	÷	Average Inventory	=	Inventory Turnover
\$358,069	÷	$\frac{\$44,858 + \$43,803}{2}$	=	8.1 times

Days in Inventory: Inventory turnover of 8.1 times divided into 365 is approximately **45.1 days**. This is the approximate time that it takes a company to sell the inventory.



© Dmitry Kutlayev/iStockphoto

Too Many TVs or Too Few?

Financial analysts closely monitored the inventory management practices of companies during the recent recession. For example, some analysts following Sony expressed concern because the company built up its inventory of televisions in an attempt to sell 25 million liquid crystal display (LCD) TVs—a 60% increase

over the prior year. A year earlier, Sony had cut its inventory levels so that its quarterly days in inventory was

down to 38 days, compared to 61 days for the same quarter a year before that. But now, as a result of its inventory build-up, days in inventory rose to 59 days. While management was saying that it didn't think that Sony's inventory levels were now too high, analysts were concerned that the company would have to engage in very heavy discounting in order to sell off its inventory. Analysts noted that the losses from discounting can be "punishing."

Source: Daisuke Wakabayashi, "Sony Pledges to Corral Inventory," Wall Street Journal Online (November 2, 2010).

For Sony, what are the advantages and disadvantages of having a low days in inventory measure? (Go to WileyPLUS for this answer and additional questions.)

Tracy Company sells three different types of home heating stoves (gas, wood, and pellet). The cost and net realizable value of its inventory of stoves are as follows.

	<u>Cost</u>	<u>Net Realizable Value</u>
Gas	\$ 84,000	\$ 79,000
Wood	250,000	280,000
Pellet	112,000	101,000

Determine the value of the company's inventory under the lower-of-cost-or-net realizable value approach.

Solution Lowest value for each inventory type is gas **\$79,000**, wood **\$250,000**, and pellet **\$101,000**. The total inventory value is the sum of these amounts, **\$430,000**.

APPENDIX 6A: Apply the inventory cost flow methods to perpetual inventory records.

Illustration

Illustration 6A-1
Inventoriable units and costs

HOUSTON ELECTRONICS Astro Condensers					
<u>Date</u>	<u>Explanation</u>	<u>Units</u>	<u>Unit Cost</u>	<u>Total Cost</u>	<u>Balance in Units</u>
1/1	Beginning inventory	100	\$10	\$ 1,000	100
4/15	Purchases	200	11	2,200	300
8/24	Purchases	300	12	3,600	600
9/10	Sale	550			50
11/27	Purchases	400	13	5,200	450
				<u>\$12,000</u>	

Assuming the **Perpetual** Inventory System, compute Cost of Goods Sold and Ending Inventory under FIFO, LIFO, and average-cost.

First-In, First-Out (FIFO)

Perpetual Inventory System

Illustration 6A-2

<u>Date</u>	<u>Purchases</u>	<u>Cost of Goods Sold</u>	<u>Balance</u> <u>(in units and cost)</u>
January 1			(100 @ \$10) \$ 1,000
April 15	(200 @ \$11)	\$2,200	
August 24	(300 @ \$12)	\$3,600	
September 10			
November 27	(400 @ \$13)	\$5,200	

Cost of Goods Sold

Ending Inventory

Last-In, First-Out (LIFO)

Perpetual Inventory System

Illustration 6A-3

<u>Date</u>	<u>Purchases</u>	<u>Cost of Goods Sold</u>	<u>Balance</u> <u>(in units and cost)</u>
January 1			(100 @ \$10) \$ 1,000
April 15	(200 @ \$11)	\$2,200	
August 24	(300 @ \$12)	\$3,600	
September 10			
November 27	(400 @ \$13)	\$5,200	

Cost of Goods Sold

Ending Inventory

Average-Cost

Moving Average Method

Illustration 6A-4

<u>Date</u>	<u>Purchases</u>	<u>Cost of Goods Sold</u>	<u>Balance</u> <u>(in units and cost)</u>	
January 1			(100 @ \$10)	\$ 1,000
April 15	(200 @ \$11) \$2,200			
August 24	(300 @ \$12) \$3,600			
September 10				
November 27	(400 @ \$13) \$5,200			

Cost of Goods Sold

Ending Inventory

Gross Profit Method

A method of estimating the cost of ending inventory by applying a gross profit rate to net sales.

A company needs to know its **net sales**, **cost of goods available for sale**, and **gross profit rate**.

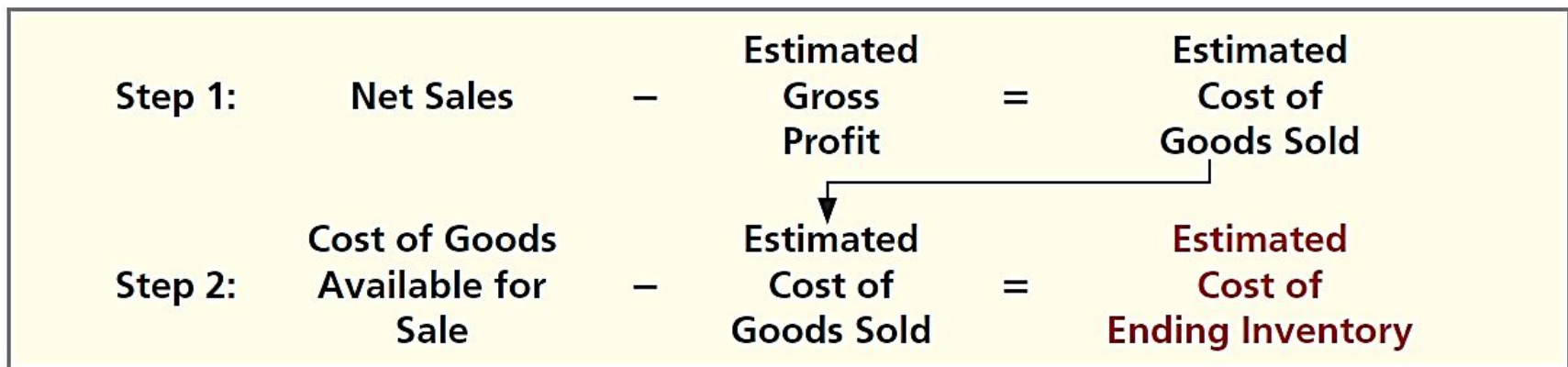


Illustration 6B-1
Gross profit method formulas

Gross Profit Method

Illustration: Kishwaukee Company records show net sales of \$200,000, beginning inventory \$40,000, and cost of goods purchased \$120,000. In the preceding year, the company realized a 30% gross profit rate. It expects to earn the same rate this year. Compute the estimated cost of the ending inventory at January 31 under the gross profit method.

Step 1:

Net sales	\$ 200,000
Less: Estimated gross profit ($30\% \times \$200,000$)	<u>60,000</u>
Estimated cost of goods sold	<u><u>\$140,000</u></u>

Step 2:

Beginning inventory	\$ 40,000
Cost of goods purchased	<u>120,000</u>
Cost of goods available for sale	160,000
Less: Estimated cost of goods sold	<u>140,000</u>
Estimated cost of ending inventory	<u><u>\$ 20,000</u></u>

Illustration 6B-2
Example of gross
profit method

Retail Inventory Method

- ▶ Retail companies establish a relationship between cost and sales price.
- ▶ Company applies cost-to-retail percentage to ending inventory at retail prices to determine inventory at cost.

Step 1:	Goods Available for Sale at Retail	–	Net Sales	=	Ending Inventory at Retail
Step 2:	Goods Available for Sale at Cost	÷	Goods Available for Sale at Retail	=	Cost-to- Retail Ratio
Step 3:	Ending Inventory at Retail	×	Cost-to- Retail Ratio	=	Estimated Cost of Ending Inventory

Illustration 6B-3

Retail inventory method formulas

Retail Inventory Method

Illustration: It is not necessary to take a physical inventory to determine the estimated cost of goods on hand at any given time.

Illustration 6B-4

	<u>At Cost</u>	<u>At Retail</u>
Beginning inventory	\$14,000	\$ 21,500
Goods purchased	<u>61,000</u>	<u>78,500</u>
Goods available for sale	<u>\$75,000</u>	100,000
Less: Net sales		<u>70,000</u>
		<u><u>\$ 30,000</u></u>

Step (1) Ending inventory at retail =

Step (2) Cost-to-retail ratio = $\$75,000 \div \$100,000 = 75\%$

Step (3) Estimated cost of ending inventory = $\$30,000 \times 75\% = \underline{\underline{\$22,500}}$

The major disadvantage of the retail method is that it is an averaging technique. It may produce an incorrect inventory valuation if the mix of the ending inventory is not representative of the mix in the goods available for sale.



A Look at IFRS

LEARNING OBJECTIVE

7

Compare the accounting for inventories under GAAP and IFRS.

Relevant Facts

Similarities

- ◆ IFRS and GAAP account for inventory acquisitions at historical cost and value inventory at the lower-of-cost-or-net-realizable value subsequent to acquisition.
- ◆ Who owns the goods—goods in transit or consigned goods—as well as the costs to include in inventory are essentially accounted for the same under IFRS and GAAP.



A Look at IFRS

Relevant Facts

Differences

- ◆ The requirements for accounting for and reporting inventories are more principles-based under IFRS. That is, GAAP provides more detailed guidelines in inventory accounting.
- ◆ A major difference between IFRS and GAAP relates to the LIFO cost flow assumption. GAAP permits the use of LIFO for inventory valuation. IFRS prohibits its use. FIFO and average-cost are the only two acceptable cost flow assumptions permitted under IFRS. Both sets of standards permit specific identification where appropriate.



A Look at IFRS

Looking to the Future

One convergence issue that will be difficult to resolve relates to the use of the **LIFO** cost flow assumption. As indicated, **IFRS** specifically prohibits its use. Conversely, the **LIFO** cost flow assumption is widely used in the United States because of its favorable tax advantages. In addition, many argue that **LIFO** from a financial reporting point of view provides a better matching of current costs against revenue and, therefore, enables companies to compute a more realistic income.



A Look at IFRS

IFRS Self-Test Questions

Which of the following should **not** be included in the inventory of a company using **IFRS**?

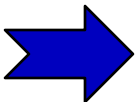
- ➡ a) Goods held on consignment from another company.
- b) Goods shipped on consignment to another company.
- c) Goods in transit from another company shipped FOB shipping point.
- d) None of the above.



A Look at IFRS

IFRS Self-Test Questions

Which method of inventory costing is prohibited under **IFRS**?

- a) Specific identification.
- b) FIFO.
-  c) LIFO.
- d) Average-cost.

Copyright

“Copyright © 2015 John Wiley & Sons, Inc. All rights reserved. Reproduction or translation of this work beyond that permitted in Section 117 of the 1976 United States Copyright Act without the express written permission of the copyright owner is unlawful. Request for further information should be addressed to the Permissions Department, John Wiley & Sons, Inc. The purchaser may make back-up copies for his/her own use only and not for distribution or resale. The Publisher assumes no responsibility for errors, omissions, or damages, caused by the use of these programs or from the use of the information contained herein.”